ARAVIND KUMAR R

AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Senior AI & Machine Learning Engineer with 7+ years of experience architecting and deploying enterprise-scale Autonomous AI Agents, Generative AI (LLMs/SLMs), NLP pipelines, and real-time Computer Vision systems. Proven track record across high-growth startups and enterprise cybersecurity environments, ranging from building agentic workflows and fine-tuning Small Language Models (SLMs) at KnowBe4, to engineering Neo4j knowledge graph recommendation engines for 1M+ learners at LearnTube.ai, and achieving 8x inference latency reductions using NVIDIA DeepStream/TensorRT at DIATOZ. Winner of KnowBe4 AI Hackathon 2026, Best Employee Award 2022, Claude Certified Architect (CCAF), and AWS Certified AI Practitioner.

TECHNICAL SKILLS

AI Agents & Generative AI: Autonomous AI Agents, Claude Agent SDK, Claude Code, Multi-Agent Systems, AI Red Teaming & Pentesting, Tool Calling / Function Calling, LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-Tuning (PEF - LoRA, QLoRA) Small Language Models (SLM), Large Language Models (LLM)

Deep Learning & Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, EmbeddingGemma, Hierarchical Multi-label Classification, Quantization (GGUF, AWQ, INT8/FP16), Vector Embeddings

Natural Language Processing (NLP): BERTopic, KeyBERT, BERT, BART, T5, LayoutLM, SpaCy, NLTK, Speech Recognition (ASR / STT), Text-to-Speech (TTS), Multi-lingual AI Dubbing, RASA, DialogFlow

Computer Vision: OpenCV, YOLO (v5/v8), NVIDIA DeepStream, Object Detection, Instance Segmentation, Real-time Video Analytics, Image Classification

MLOps & High-Performance Serving: vLLM, NVIDIA TensorRT, Triton Inference Server, Docker, FastAPI, Flask, Linux, CI/CD, Streamlit, GPU Acceleration

Cloud & Distributed Data: AWS (SageMaker, Lambda, EC2, S3, CloudWatch), Google Cloud Platform (GCP), Neo4j (Graph DB), Vector Databases (Pinecone, Chroma, Milvus), PostgreSQL, MySQL, MongoDB, Elasticsearch

Programming Languages: Python, C++, SQL, Bash

CERTIFICATIONS

- **Claude Certified Architect - Foundations (CCAF)** - Aug 2026
- **AWS Certified AI Practitioner** - Amazon Web Services (Active / July 2026)
- **AWS Certified Cloud Practitioner** - Amazon Web Services (Active / June 2025)
- **Extensive and Re-imagined AI Program v3** - The School of AI (2024 - 2025)
- **Generative AI with Large Language Models** - DeepLearning.AI & AWS (Nov 2023)
- **Career Essentials in Generative AI** - Microsoft & LinkedIn (July 2023)

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer | KnowBe4 — Jan 2026 - Present

Machine Learning Engineer | KnowBe4 — Feb 2024 - Jan 2026

Bengaluru, Karnataka, India • Total Tenure: Feb 2024 - Present (2 yrs 7 mos)

- **AIDA (Artificial Intelligence Defense Agents):** Architected and deployed agentic capabilities within AIDA, building automated phishing template generation for adaptive phishing simulations and personalized training recommendations driven by employee **RiskScore** metrics.
- **Agent Security & Prompt Injection Defense:** Engineered a prompt injection classifier based on **EmbeddingGemma** with custom classification heads, optimized inference pipelines using **vLLM** and GPU acceleration, boosting inference throughput by **6x** with low latency.
- **LangChain LLM Workflows:** Designed and deployed LangChain-based production pipelines for automated quiz generation from compliance policy documents and course modules, and built a multi-level **hierarchical multi-label, multi-class classification engine (124 classes)** for training module categorization and content discovery.
- **Autonomous Red Teaming Agent (Hackathon Winner Team):** Built an autonomous Red Teaming Agent that systematically pentests other AI agents to uncover prompt injection flaws, jailbreak vulnerabilities, and guardrail bypasses. Our team awarded **1st Place (Winner)** in the KnowBe4 AI Hackathon Bengaluru 2026.
- **Agentic Workflows & Safety:** Spearheaded agentic evaluation pipelines and developer automation utilizing the **Claude Agent SDK** and **Claude Code** for automated agent auditing and internal security testing.

Machine Learning Engineer | LearnTube.ai — Oct 2021 - Aug 2023 (1 yr 11 mos)

CareerNinja (BeNinja Solutions)

- Spearheaded core AI/ML systems powering an AI-driven learning platform serving **over 1,000,000 learners**, reducing end-to-end curriculum and content generation cycles from days to under 2 hours.
- Engineered, deployed, and scaled a **Neo4j Knowledge Graph** recommendation system connecting skills, learning objectives, and course videos to deliver dynamic personalized learning pathways.
- Built automated AI video dubbing pipeline orchestrating Whisper speech-to-text (ASR), neural machine translation, and multilingual voice synthesis to localize educational video content into regional Indian languages.
- Architected an NLP discovery engine leveraging **BERTopic, KeyBERT, and automated web crawlers** to ingest, topic-model, and rank educational YouTube videos with sub-second retrieval latency.
- Developed production Generative AI features including Instant Doubt Solving, AI Question Generation, and Summarization using GPT-4, Claude, LLaMA, and open-source foundation models.
- Awarded the **Best Employee Award (2022)** for instrumental engineering leadership and delivery of flagship GenAI capabilities.

PROFESSIONAL EXPERIENCE (CONTINUED)

Senior Software Development Engineer - Applied AI | DIATOZ Solutions — Sep 2020 - Oct 2021

Software Development Engineer - Applied AI | DIATOZ Solutions — Aug 2019 - Sep 2020

Bengaluru, Karnataka, India • Total Tenure: Aug 2019 - Oct 2021 (2 yrs 3 mos)

- Led an engineering team of 4 in designing, deploying, and maintaining AI engines for **E2Ehiring** (e2ehiring.com), an AI recruitment platform featuring automated resume information extraction (LayoutLM / OCR), candidate-job matching algorithms, and automated interview assessment scoring.
- Delivered an **8x inference speedup** across real-time video analytics pipelines by migrating deep learning models to **NVIDIA DeepStream, TensorRT, and Triton Inference Server** with FP16/INT8 model quantization.
- Led a 3-engineer squad to build and maintain **Ezevision**, a SaaS traffic analytics platform providing real-time multi-camera vehicle detection, speed estimation, and congestion monitoring.
- Built multi-lingual customer support conversational agents supporting 4 Indian languages for a tier-1 Indian infrastructure enterprise utilizing RASA, DialogFlow, and custom BERT intent classifiers.
- Authored production deployment standards and mentored engineers on containerized microservice architectures using Docker, FastAPI, and GPU acceleration.

Project Management Trainee - GAT | Emerson Automation Solutions — Mar 2018 - Oct 2018 (8 mos)

Chennai, Tamil Nadu, India

- Assisted Project Manager in delivering the facility's first advanced High-Integrity Pressure Protection System (HIPPS), coordinating technical specifications, quality requirements, and cross-functional expert reviews.

HONORS & AWARDS

- **Winner - KnowBe4 AI Hackathon Bengaluru (2026):** Built an autonomous Red Team Hacking Agent evaluating AI agent security, prompt injection defenses, and vulnerability assessment.
- **Runners Up - Engineering Innovation Summit (2024):** Recognized for excellence in architecting and deploying real-world generative AI and agentic systems.
- **Best Employee Award (2022) - LearnTube.ai / CareerNinja:** Conferred for outstanding impact in scaling GenAI features and Neo4j graph recommendations to 1M+ users.
- **Best Paper Award (2017) - IEEE Sponsored ICRAET:** Primary author of research paper on *"AURATER: Autonomous Unmanned Rover with Advanced Trash Eliminating Robot"*.
- **Achieving Student Award (2017):** Conferred by Sri Krishna College of Engineering and Technology for academic and technological innovation.

PUBLICATIONS

- **AURATER: Autonomous Unmanned Rover with Advanced Trash Eliminating Robot** *International Conference on Recent Advances in Engineering and Technology (ICRAET - IEEE Sponsored)*

EDUCATION

Bachelor of Engineering (B.E.) in Mechatronics Engineering — 2015 - 2018

Sri Krishna College of Engineering and Technology, Coimbatore, Tamil Nadu, India • **CGPA: 8.37** / 10
(NIRF Top 100 Engineering Institution)

Diploma in Mechanical Engineering — 2012 - 2015

Nachimuthu Polytechnic College, Coimbatore, Tamil Nadu, India • **Cumulative Percentage: 88%**